

NEW RDSP MODE I

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Time Limit: 1 s Memory Limit: 32 MB
Submission: 163 AC: 93 Score: 98.59

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Codes

AI Code Tips

Description

Little A has become fascinated with the game Dota recently, but he is not a good player. In all the modes, the rdsp Mode is popular on online, in this mode, little A always loses games if he gets strange heroes, because, the heroes are distributed randomly.

Little A wants to win the game, so he cracks the code of the rdsp mode with his talent on programming. The following description is about the rdsp mode:

There are N heroes in the game, and they all have a unique number between 1 and N . At the beginning of game, all heroes will be sorted by the number in ascending order. So, all heroes form a sequence One.

These heroes will be operated by the following stages M times:

1. Get out the heroes in odd position of sequence One to form a new sequence Two;
2. Let the remaining heroes in even position to form a new sequence Three;
3. Add the sequence Two to the back of sequence Three to form a new sequence One.

After M times' operation, the X heroes in the front of new sequence One will be chosen to be Little A's heroes. The problem for you is to tell little A the numbers of his heroes.

Input

There are several test cases.

Each case contains three integers N ($1 \leq N < 1,000,000$), M ($1 \leq M < 100,000,000$), X ($1 \leq X \leq 20$).

Proceed to the end of file.

Output

For each test case, output X integers indicate the number of heroes. There is a space between two numbers. The output of one test case occupied exactly one line.

Samples

input	Copy
5 1 2 5 2 2	
output	Copy
2 4 4 3	

Hint

In case two: $N=5, M=2, X=2$, the initial sequence One is 1,2,3,4,5. After the first operation, the sequence One is 2,4,1,3,5. After the second operation, the sequence One is 4,3,2,1,5. So,output 4 3.

Source

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